

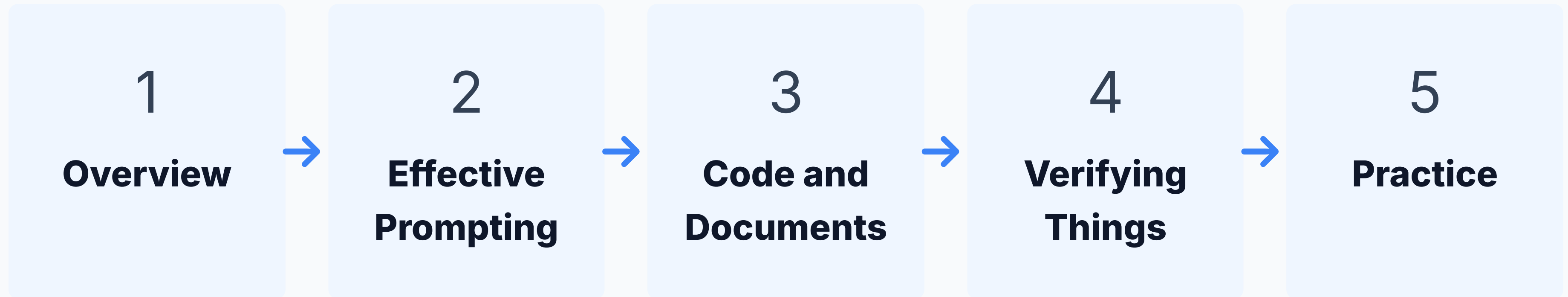
1. Intro to Claude

Kerry Back

Faculty Workshop

Rice Business 2026

Today's Session

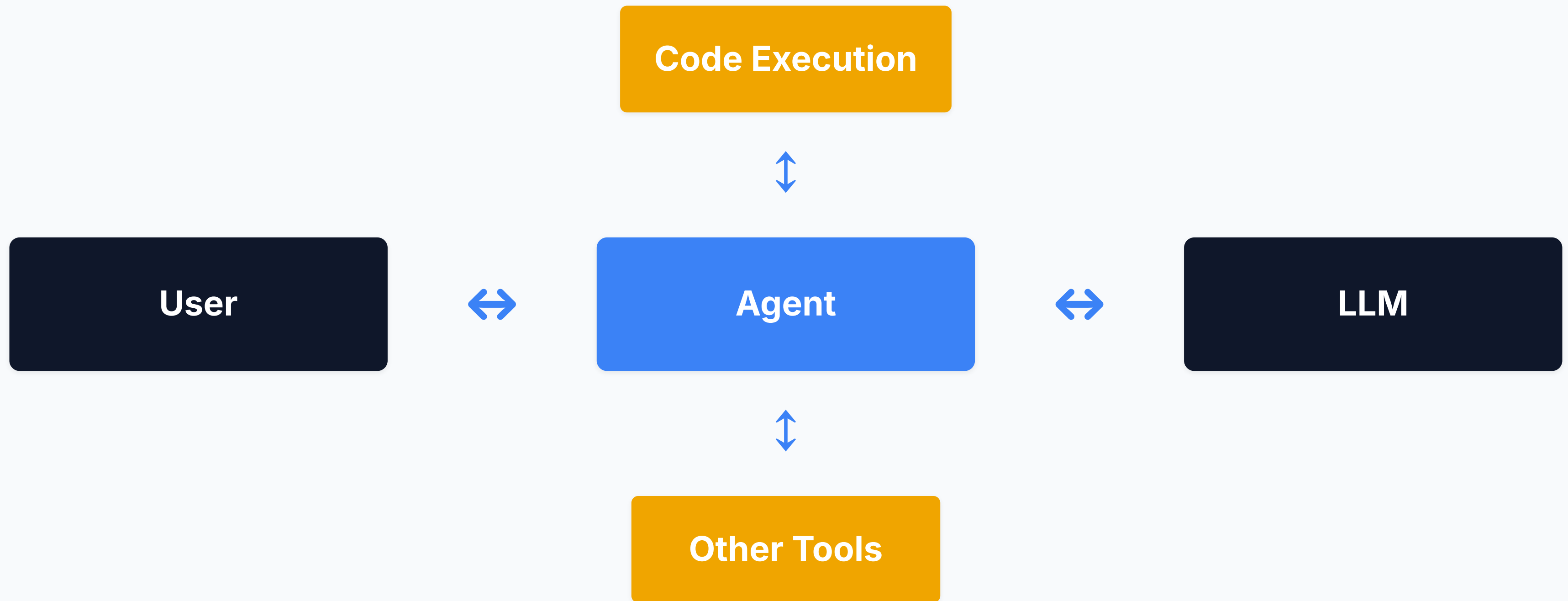


Overview

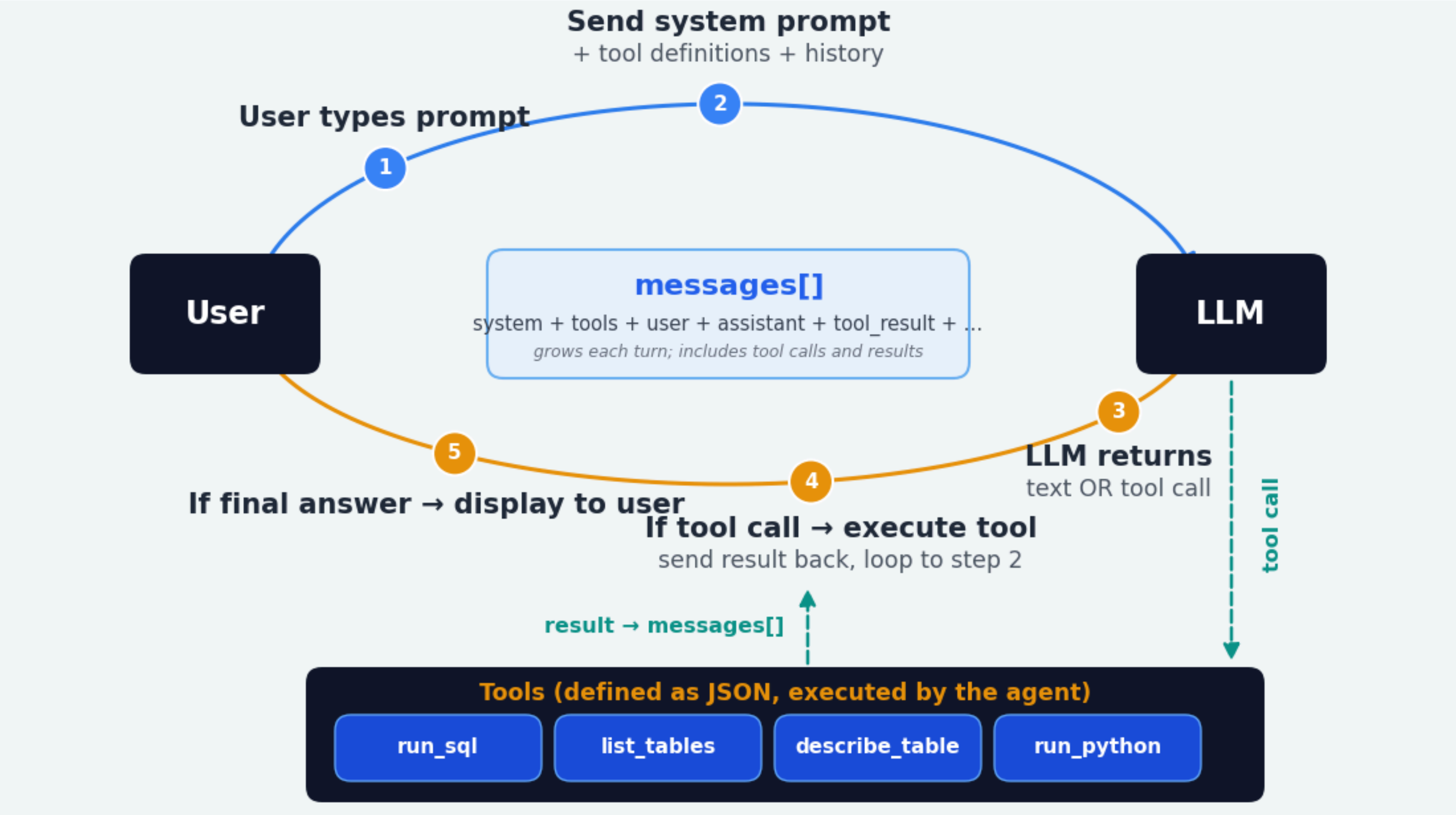
A Brief History



An Agent is a Chatbot with Tools



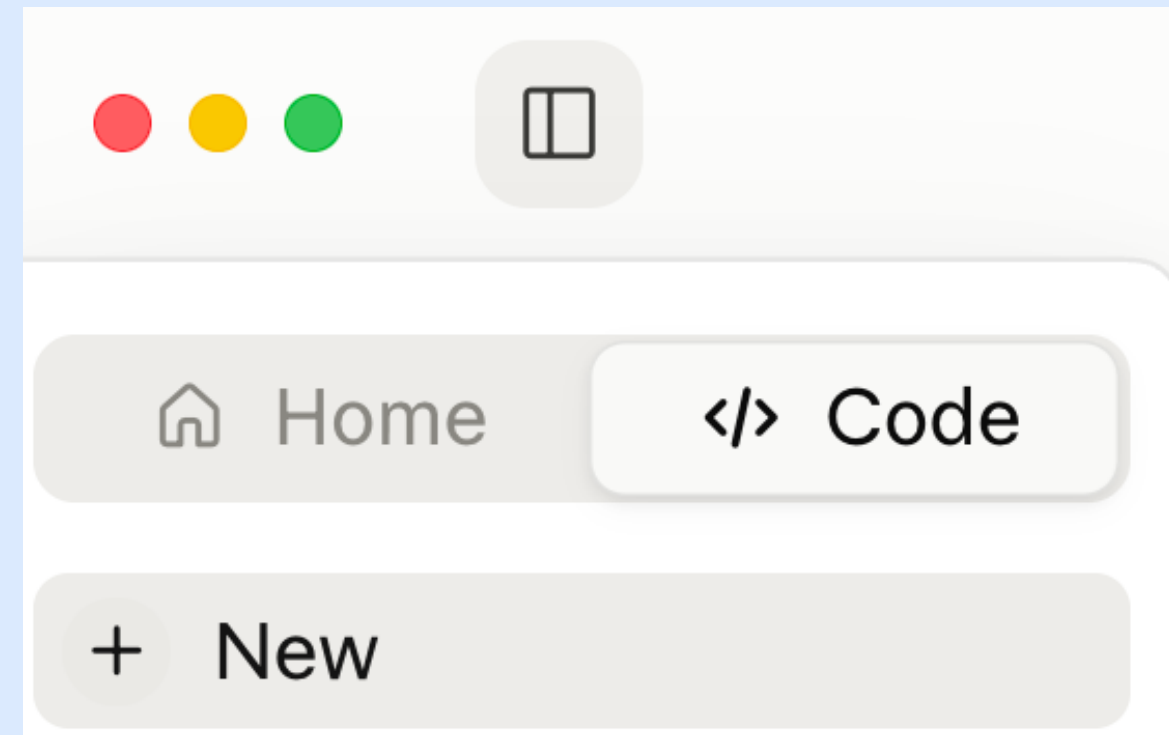
Agent in More Detail



Four Modes of Claude Desktop

Home or Code?

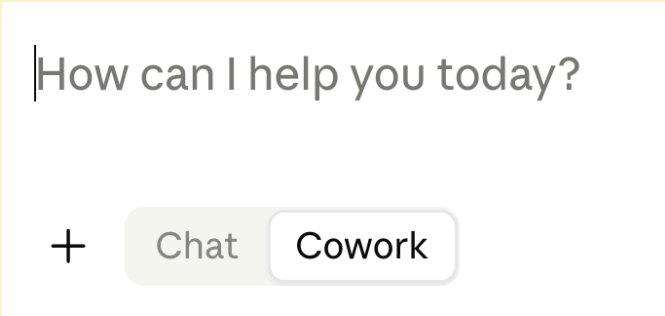
The two halves of the app, chosen at the top of the left sidebar (the square icon toggles the sidebar)



Four Modes of Claude Desktop

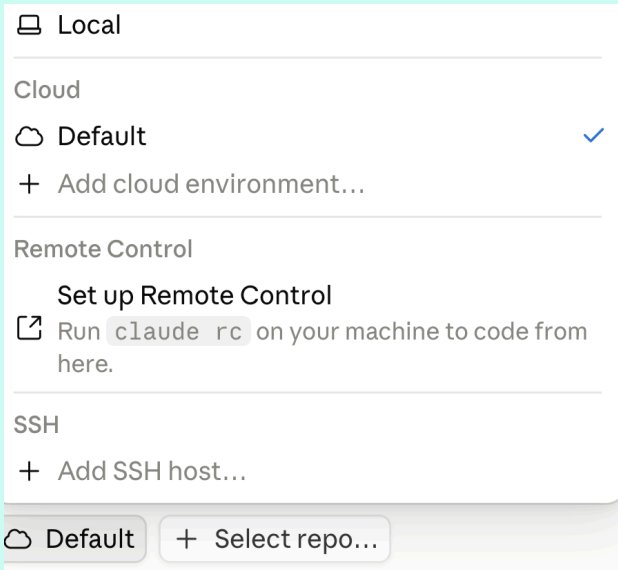
Within Home, Chat or Cowork?

Toggle in the prompt box



Within Code, Local or Cloud?

Selector at the bottom of the window



What to Use?

- You can chat in any mode
- You can execute code in any mode

Cowork > Chat on many dimensions.
Requires paid account.

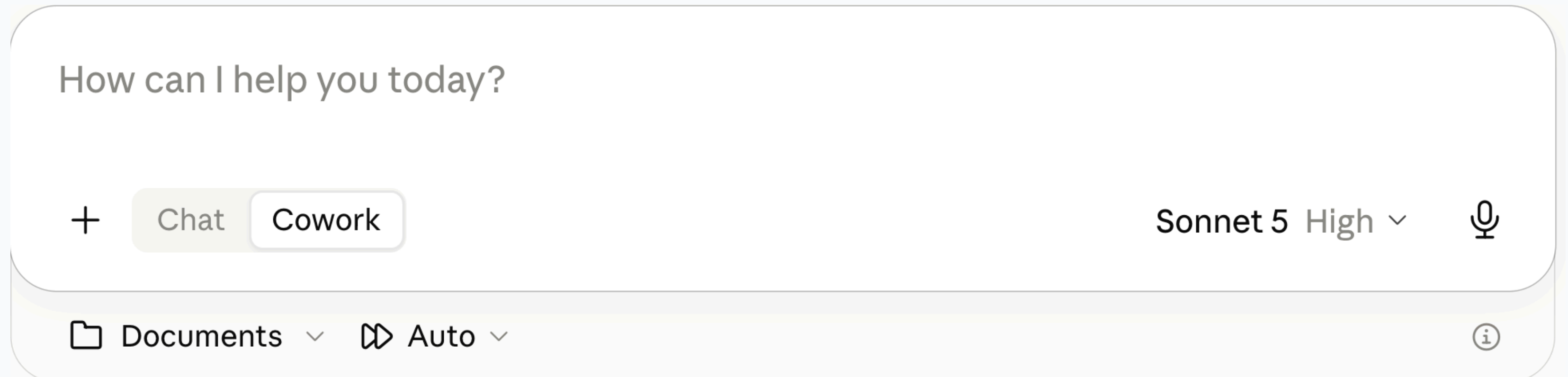
Code Local is best for most people, but Claude Desktop is not the best venue

Use Cowork for now
(more later)

Code Cloud is GitHub integration, probably overkill

Configuration

- **Create New Project + Use a Folder** Give Claude access to your local files. Store project-specific instructions.
- **Auto** Automatically approve. Misleading name. Claude decides: safe → don't ask permission, possibly risky → ask permission
- **Model** I recommend Sonnet 5. Capable for anything you probably want to do. Faster and cheaper (easier on token quota).



How can I help you today?

+ Chat Cowork

Sonnet 5 High ▾

🎤

📁 Documents ▾ 🎞️ Auto ▾ ⓘ

The screenshot shows the Claude AI interface. At the top is a text input field with the placeholder text "How can I help you today?". Below the input field are two buttons: "Chat" (highlighted in light green) and "Cowork" (in a light grey box). To the right of these buttons, the model is set to "Sonnet 5" and the temperature to "High", both with a dropdown arrow. A microphone icon is to the right of the temperature setting. At the bottom of the interface, there are two more dropdown menus: "Documents" (with a folder icon) and "Auto" (with a play icon). An information icon (i in a circle) is located at the bottom right of the interface.

Effective Prompting

Plan Mode

Claude Desktop has an explicit Plan mode. Or add “plan how to do this before you start.”

Why It Works: Knowledge

- Planning pulls relevant knowledge into the context window
- The model answers in light of what it just retrieved, not base rates

Why It Works: Scale

- Decompose the work into subtasks
- Verify each one before the next
- Failures stay small and local

AI is a Prediction Machine, not a Rational Person

Suppose Claude has told you in the past that Method A beats Method B for some task. You ask it to do the task.

Do not assume it will use what it knows. If "A beats B" is not in the context window, and B appeared more often in its training data, it will probably use B.

This is the value of "plan before you act." Planning surfaces "A beats B," leading Claude to select A for the task.

Steelmanning

Don't ask AI whether your idea is good. Ask it to make the strongest possible case that your idea is **wrong**.

Sycophancy: training on human feedback teaches that agreement gets high ratings. Ask "is my plan sound?" and Claude finds reasons it is.

The fix: "What is the strongest argument against this acquisition?"

One step further: have it play a skeptical board member, a regulator, a competitor.

Numbers Need Code

Never trust AI arithmetic. Have it write and run code.

The Problem

- "47.3%" is as plausible as "52.1%" to a next-token predictor
- Mental math, rankings, comparisons, aggregations — all unreliable
- Wrong numbers stated with complete confidence

The Fix

- Ask for Python, and have it run
- Arithmetic becomes computation, not prediction
- Code execution is built into every chatbot and Claude Code

What Not to Do

2022–2023 conventions are now useless or counterproductive.

Skip These

- “Think step by step” — reasoning models already do, in hidden tokens
- “Take a deep breath” — measurably useless now
- “CRITICAL! NEVER EVER...” — overtriggers, and gets worse results

Use With Care

- “You are an expert data analyst ...”
- Persona stacking can improve tone and structure
- Direct task framing usually wins

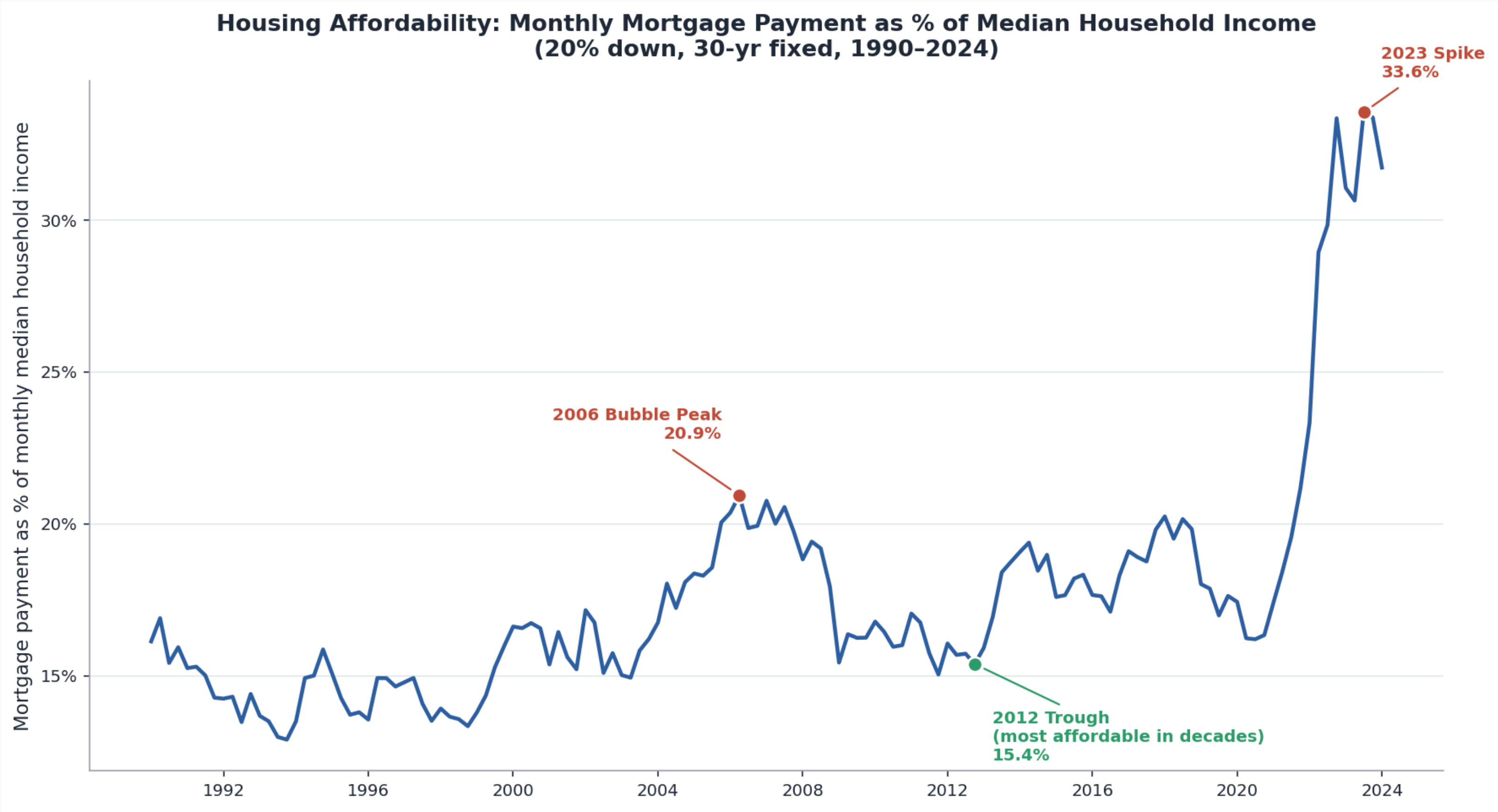
Code and Documents

Example Prompt

Download three FRED (Federal Reserve Economic Data) series as CSV: mortgage rate (MORTGAGE30US), home price (MSPUS), household income (MEH0INUSA672N), 1990 through most recent.

1. MORTGAGE30US is weekly – resample to quarterly by averaging. MSPUS is already quarterly. MEH0INUSA672N is annual – interpolate linearly between annual observations to produce quarterly values.
2. Assume a 20% down payment and a 30-year loan. Compute the monthly mortgage payment on a median-priced home and express it as a percentage of monthly median household income.
3. Plot the affordability ratio as a line chart. Annotate three points: the 2006 bubble peak, the 2012 trough (most affordable in decades, as low rates met depressed prices), and the 2023 spike.

Result



Reading, Editing, and Writing Documents

Claude natively reads **text and images**, and natively writes **text**. Everything beyond text is handled by code.

- Document reading is doc → text, then read
- Document creation is text (code) → doc
- Office document editing is 'read and write XML (Extensible Markup Language)'

Example

Get Tesla's EPS from SEC EDGAR by quarter for the past 5 years. Plot. Create a PowerPoint deck with the plot plus discussion.

The result

[tesla.pptx](#)

Creating Excel Workbooks

Claude uses Python to create Excel workbooks, inserting numbers (inputs) and formulas cell-by-cell

Example Prompt

```
Build an Excel workbook containing a mortgage amortization table. Include a chart of the amount going to interest and the amount going to principal each month.
```

Result

[amortization.xlsx](#)

Claude Plugin for Excel

A chat panel inside Excel itself. Add it from Home → Add-ins. You will be prompted to connect to your Anthropic account (paid account).

Example

	A	B	C	D	E	F	G	H	I	J	K
1	Mortgage Amortization Schedule										
2											
3	Inputs										
4	Loan Amount	\$400,000									
5	Annual Interest	6.500%									
6	Term (Years)	30									
7	Payments per	12									
8	Start Date	01/01/2026									
9											
10	Outputs										
11	Total Payment	360									
12	Periodic Rate	0.5417%									
13	Monthly Paym	\$2,528.27									
14	Total Paid	\$910,177.95									
15	Total Interest	\$510,177.95									
16											
17	Amortization Schedule										
18	Period	Date	Beginning Balance	Payment	Principal	Interest	Ending Balance	Cumulative Interest			
19	1	01/01/2026	\$400,000.00	\$2,528.27	\$361.61	\$2,166.67	\$399,638.39	\$2,166.67			
20	2	02/01/2026	\$399,638.39	\$2,528.27	\$363.56	\$2,164.71	\$399,274.83	\$4,331.37			
21	3	03/01/2026	\$399,274.83	\$2,528.27	\$365.53	\$2,162.74	\$398,909.30	\$6,494.11			
22	4	04/01/2026	\$398,909.30	\$2,528.27	\$367.51	\$2,160.76	\$398,541.78	\$8,654.87			
23	5	05/01/2026	\$398,541.78	\$2,528.27	\$369.50	\$2,158.77	\$398,172.28	\$10,813.64			
24	6	06/01/2026	\$398,172.28	\$2,528.27	\$371.51	\$2,156.77	\$397,800.77	\$12,970.41			
25	7	07/01/2026	\$397,800.77	\$2,528.27	\$373.52	\$2,154.75	\$397,427.26	\$15,125.16			
26	8	08/01/2026	\$397,427.26	\$2,528.27	\$375.54	\$2,152.73	\$397,051.71	\$17,277.89			
27	9	09/01/2026	\$397,051.71	\$2,528.27	\$377.58	\$2,150.70	\$396,674.14	\$19,428.59			
28	10	10/01/2026	\$396,674.14	\$2,528.27	\$379.62	\$2,148.65	\$396,294.52	\$21,577.24			
29	11	11/01/2026	\$396,294.52	\$2,528.27	\$381.68	\$2,146.60	\$395,912.84	\$23,723.84			

Claude

⌚ ⊕ ⋮ i

create a mortgage amortization table. assume a 30-year mortgage. allow the other parameters to be inputs.

I'll build a mortgage amortization table on Sheet1 — an inputs block (loan amount, rate, etc.) feeding a 360-month schedule for the 30-year term.

Let me set up the inputs section first.

Build inputs block in A1:B8 >

Now the calculated summary outputs:

Build outputs block in A10:B15 >

Payment of \$2,528.27/mo checks out. Now the schedule headers and the first row, then I'll fill down 360 periods.

Verifying Things

Verify Claims

The Problem

- Facts, regulations, dates, citations — stated the same way whether real or invented
- The more obscure the fact, the likelier it is fabricated

The Fix

- Have it search before answering
- Ask for sources. Then check them.

Verify Code

Consider a human assistant. Ask where they could have gone wrong.

- They may not have understood what you wanted.
- To perform the task, they may have had to make decisions that were different from the decisions you would have made.
- Their Excel, code, or whatever may have had subtle bugs so it did not do what they expected.

Claude can go wrong in exactly the same ways. How do you check them?

Three Checks

Take advantage of the fact that **AI is perfectly patient**. You can probe more deeply than you would with a human assistant, and it will not be offended.

Check its understanding

Explain to me what you did and how you did it.



Check its decisions

Report all ways in which my instructions were ambiguous and you made decisions we have not discussed.



Check its code

Show me a sample case and work through the case as if by hand so I can verify your number.

Generator-Critic

Spawn a second Claude to attack the first one's output, not validate it. Subagents work well — several of them.

The Pattern

- **Generator** produces the work
- **Critic** gets: "Find every factual error, logical gap, and unsupported assumption. Do not summarize — find flaws."

Why a Separate Instance?

- A model is anchored to its own output and defends it
- A fresh instance with an adversarial mandate is not

Re-Use Tested Code

Once code runs correctly, save it. Re-use it rather than rewrite it.

The Problem with Rewriting

- Different choices each time
- Scripts that should agree may not
- Old bugs reappear

Practice

Some Local Data

Extract the following from session1.zip into your project folder or just put session1.zip there and tell Claude to extract.

File	Rows	Key Columns
northwind_customers.xlsx	93	CustomerID, CompanyName, Country
northwind_orders.xlsx	500	OrderID, CustomerID, OrderDate
northwind_orderdetails.xlsx	18,976	OrderID, ProductID, UnitPrice, Quantity
northwind_products.xlsx	77	ProductID, ProductName, CategoryID
northwind_categories.xlsx	8	CategoryID, CategoryName

Example: Merge, Filter, Aggregate, Chart, Analyze, Save

Copy this prompt into Claude:

```
I have five Northwind files in my project folder: customers, orders, order details, products, and categories. Which product categories generated the most revenue from customers in Germany and France? Show only categories above $10,000, sort highest to lowest, create a bar chart, and put it in a PowerPoint deck. Add some discussion.
```

Here we put a lot into one prompt. More commonly, we would ask it to take first steps, then do something else, then do something else, ... Either works fine.

Homework

Watch the three-minute video [academic_studio.mp4](#)